Graph Coloring

CS1200: Introduction to Algorithms and their Limitations
Lecture 12

Fall 2026

Harvard SEAS

Put away your devices

Announcements

- SRE 2 (Beating Exhaustive Search for Coloring): most of today.
- Pset 4 due today.
- Reading: Hesterberg–Vadhan §13.2, §14.1–14.2; Senders: Appendix A.6.

Recap: coloring and greedy

Recap: coloring, and one greedy run

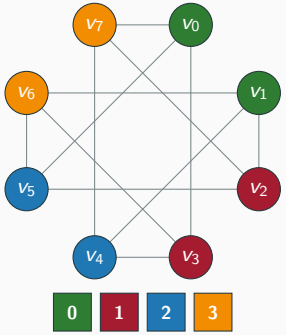
Def. A k-coloring of $G = (V, E)$ is $f : V \to \{0, \ldots, k-1\}$ with $f(u) \neq f(v)$ for every $\{u, v\} \in E$.

Graph Coloring: given G and k, output a k-coloring, or $\perp$ if none exists.

GreedyColoring(G): fix an order $v_0, \ldots, v_{n-1}$; give each v_i the smallest color not already on a neighbor.

- Always a proper coloring. Time $O(n + m)$.
- Clockwise order on the graph at right: 4 colors.

Is 4 the best possible for this graph?



How many colors does greedy use?

Theorem 13.3: greedy never does worse than $\deg_{\max} + 1$

Theorem 13.3. For any graph G and any vertex order, GreedyColoring(G) uses at most $\deg_{\max} + 1$ colors, where $\deg_{\max} = \max_v \deg(v)$.

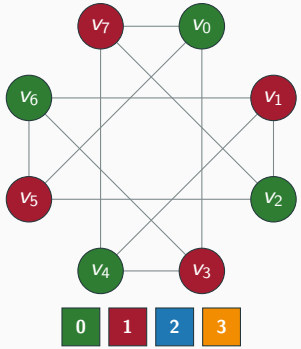
Proof. When v_i is colored, the colors already on its neighbors form a set of size at most $\deg(v_i) \leq \deg_{\max}$. So at least one of the colors $0, 1, \ldots, \deg_{\max}$ is free, and greedy picks the smallest free one. □

Remarks.

- An algorithmic proof of a graph-theory fact: every graph is $(\deg_{\max} + 1)$-colorable.
- The bound can be far from the truth. Our graph: $\deg_{\max} + 1 = 4$, and greedy hit it.

A better order

In groups: greedy in BFS order



BFSColoring(G)**:** run BFS from v_0; greedy-color in the order BFS discovers vertices (frontier by frontier).

1. Run BFS from v_0. What are the frontiers?
   $F_0 = \{v_0\}$, $F_1 = \{v_3, v_5, v_7\}$,
   $F_2 = \{v_2, v_4, v_6\}$, $F_3 = \{v_1\}$
2. Greedy-color in that order. How many colors?
   Two: F_0, F_2 get 0; F_1, F_3 get 1.
3. Why did it work?
   Every edge joins consecutive frontiers; no edge inside a frontier.

Why BFS order works: Theorem 13.4

Theorem 13.4. If G is connected and 2-colorable, then BFSColoring(G) uses exactly 2 colors.

Def. G is bipartite if V splits into parts W, W' with every edge having one endpoint in each.

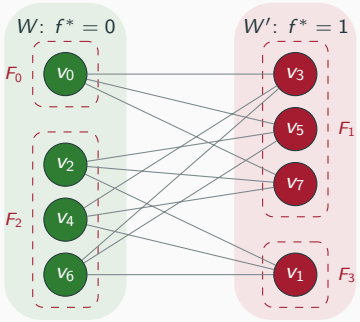
Fact. 2-colorable $\iff$ bipartite (proof in appendix).

Proof idea. Fix a 2-coloring f^*, i.e. a bipartition $W = \{v : f^*(v) = 0\} \ni v_0$, $W' = \{v : f^*(v) = 1\}$.

Claim $(*)$**.** For every d:

- $F_d \subseteq W$ for even d, $F_d \subseteq W'$ for odd d;
- greedy gives color 0 on even F_d, color 1 on odd F_d.

Induction on d. Each vertex of F_{d+1} has a neighbor in F_d, so it is on the other side. Its colored neighbors all lie in F_d, all of one color, so greedy picks the other. $\square$



Corollary 13.5: Graph 2-Coloring in $O(n + m)$

Corollary 13.5. Graph 2-Coloring can be solved in time $O(n + m)$.

Algorithm.

1. Split G into connected components: BFS from any vertex; then BFS from any undiscovered vertex; repeat.
2. Run BFSColoring on each component.
3. If every component got 2 colors, return f; otherwise return $\perp$ — by Thm 13.4, no 2-coloring exists.

Total time: each BFS visits each vertex and edge once, so $O(n + m)$ overall.

From 2 to 3

Two colors, three colors

problem	best algorithm so far	time
GRAPH 2-COLORING	BFSColoring per component	$O(n+m)$
GRAPH 3-COLORING	exhaustive search: try all 3^n assignments	$O(m \cdot 3^n)$

Today's SRE (Theorem 14.2). GRAPH 3-COLORING can be solved in time $O(1.89^n)$.

- Still exponential. But $n = 60$: $3^{60} \approx 4 \cdot 10^{28}$ vs. $1.89^{60} \approx 4 \cdot 10^{16}$.
- Best known: about 1.33^n. No polynomial-time algorithm is known.

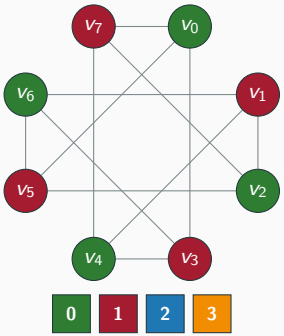
Independent sets (§14.1)

Def. 14.1. An independent set in $G = (V, E)$ is a set $S \subseteq V$ with no edge inside it: for every $\{u, v\} \in E$, $u \notin S$ or $v \notin S$.

In the graph at right:
$\{v_0, v_2\}$: independent $\quad \{v_0, v_3\}$: not $(\{v_0, v_3\} \in E)$
$\{v_0, v_2, v_4, v_6\}$: independent

Key fact. A k-coloring of G is the same thing as a partition of V into k independent sets (one per color: a color class has no edge inside it, and conversely).

Independent Set: given G, find a largest independent set. Wednesday's lecture (Ch 14).



SRE 2

Sender–Receiver Exercise 2: beating exhaustive search for coloring

Theorem 14.2. Graph 3-Coloring can be solved in time $O(1.89^n)$.

- **Pair up:** one Sender, one Receiver. No partner? Raise your hand; a group of three is fine.
- **Sender:** explain the theorem and proof, top-down — the main idea first, then the details. Avoid writing; one diagram to get started is fine.
- **Receiver:** take notes and draw pictures, ask questions, and fill in as many formal details as you can.
- **Stuck?** Raise your hand; staff will come.
- **At the end:** fill out the reflection survey.

Takeaway

- Greedy always colors properly, with at most $\deg_{\max} + 1$ colors. The order is everything.
- BFS order is provably optimal for 2-colorable (= bipartite) graphs: 2-Coloring in $O(n + m)$.
- 3-Coloring: no efficient algorithm known. Today's SRE: exhaustive search over the smallest color class ($|S| \leq n/3$), then 2-color the rest: $O(1.89^n)$ instead of $O(m \cdot 3^n)$. Where 1.89 comes from: appendix. Best known: about 1.33^n.
- A coloring is a partition into independent sets. Wednesday: Independent Set and Interval Scheduling (Ch 14, Anurag). Monday 10/26: Matching (Ch 15).

Appendix: the 1.89 algorithm

The lemma behind the algorithm

Write $G_{\setminus S}$ for G with the vertices of S (and their edges) deleted.

Lemma. For a graph $G = (V, E)$ with n vertices:

1. If G is 3-colorable, there is an independent set S with $|S| \leq n/3$ such that $G_{\setminus S}$ is 2-colorable.
2. If S is independent and $G_{\setminus S}$ has a 2-coloring $f_{\setminus S}$, then G is 3-colorable: $f(v) = f_{\setminus S}(v)$ for $v \notin S$, and $f(v) = 2$ for $v \in S$.

Why 1. Take a 3-coloring. Its three color classes partition V, so the smallest has size $\leq n/3$. Call it S: it is independent, and the other two classes 2-color $G_{\setminus S}$.

Why 2. Check each edge $\{u, v\}$: both ends in S is impossible (S independent); one end in S gets 2 vs. 0 or 1; neither end in S is an edge of $G_{\setminus S}$, properly 2-colored.

The algorithm and its runtime

3by2Coloring(G)**:**

0 **foreach** $S \subseteq V$ with $|S| \leq n/3$:
1 **if** S is an independent set:
2 construct $G_{\setminus S}$
3 $f_{\setminus S} = 2\text{-}\textsc{Coloring}(G_{\setminus S})$
4 **if** $f_{\setminus S} \neq \perp$:
5 **return** f as in the lemma
6 **return** $\perp$

Per candidate S**:** lines 1–5 take $O(n+m)$.
(scan the edges; Cor. 13.5.)

Number of candidates: subsets of size $\leq n/3$:

$$\sum_{k \leq n/3} \binom{n}{k} \;\leq\; c^n, \quad c = 3^{1/3} \cdot (3/2)^{2/3} \approx 1.8899$$

Total: $O\big((n+m) \cdot c^n\big) = O(1.89^n)$
since $n + m \leq n^2 \leq (1.89/c)^n$ for large n.

Appendix: 2-colorable $\iff$ bipartite

Appendix: 2-colorable $\Rightarrow$ bipartite

Def. $G = (V, E)$ is bipartite if there are $W, W' \subseteq V$ with $W \cup W' = V$, $W \cap W' = \emptyset$, and every edge $\{u, v\} \in E$ has one endpoint in W and the other in W'.

Claim. If G has a 2-coloring, then G is bipartite.

Proof. Let $f : V \to \{0, 1\}$ be a 2-coloring. Set $W = \{v : f(v) = 0\}$ and $W' = \{v : f(v) = 1\}$.

- *Partition.* Every v has exactly one color, so v lies in exactly one of W, W'.
- *Edges cross.* For $\{u, v\} \in E$, $f(u) \neq f(v)$, so one of $f(u), f(v)$ is 0 and the other is 1: one endpoint in W, the other in W'. $\square$

On the activity graph: $f =$ the BFS coloring, $W = \{v_0, v_2, v_4, v_6\}$, $W' = \{v_1, v_3, v_5, v_7\}$.

Appendix: bipartite $\Rightarrow$ 2-colorable

Claim. If G is bipartite with parts W, W', then

$$f(v) = \begin{cases} 0 & v \in W \\ 1 & v \in W' \end{cases}$$

is a 2-coloring of G.

Proof.

- *Well defined.* W, W' partition V, so every v gets exactly one value.
- *Proper.* For $\{u, v\} \in E$, one endpoint is in W and the other in W', so $\{f(u), f(v)\} = \{0, 1\}$ and $f(u) \neq f(v)$. $\square$

So: 2-colorable $\iff$ bipartite, and the bipartition *is* the coloring. Theorem 13.4 says BFS finds it whenever it exists; Corollary 13.5 makes that an $O(n + m)$ test for bipartiteness.